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Cooperative ISAC: An End-to-End Perspective

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(Invited Article)

ABSTRACT Integrated sensing and communication (ISAC) has caught a great deal of interest and attention since it was identified as a key usage scenario of sixth generation (6G) mobile communications. Whereas cooperative ISAC, taking advantage of the large-scale cellular infrastructure, in addition to the benefit of eliminating full duplex transceivers, is favored by mobile operators. This article starts with channel modeling and key technologies at the link level, including synchronization, beam management, and advanced sensing algorithms, of cooperative ISAC systems. From a network perspective, it provides optimal base station (BS) deployment, receiving node(s) selection, and network interference management to improve system performance. Then, cooperative ISAC enhanced by novel multiple-input multiple-output (MIMO) antenna technologies is introduced. Finally, prototype designs of cooperative ISAC systems at both sub-7 GHz and millimeter wave bands are presented, respectively. Experimental results in an outdoor environment have verified that with only 3.5% of the communication capacity cost and 200 MHz bandwidth, a range estimation accuracy of 0.2 m can be achieved within the 500 m range.

INDEX TERMS Cooperative, integrated sensing and communication (ISAC), multiple-input multiple-output (MIMO), prototype.

INTRODUCTION

Integrated sensing and communication (ISAC) is without a doubt one of the two most intriguing usage scenarios among the six key usage scenarios officially targeted in the sixth generation (6G) mobile network. ISAC will empower novel applications including guidance, monitoring and regulation of unmanned aerial vehicles (UAVs) in the digital low-altitude economy, as well as vehicle-to-everything (V2X) in intelligent transportation and city management [1], [2]. It combines the radar sensing and wireless communication into a single system with maximally shared hardware and spectrum resources [3]. This combination also brings mutual benefits to both communication and sensing functions (SFs). Specifically, the SF may obtain parameters of channel scatters, including angle, range, and velocity information [4]. These parameters provide additional insight into the propagation environment, which can be used to enhance the channel estimation and reconstruction in communication. While the communication function is the fundamental basis of the sensing since it carries out the task of sensing information delivery and signaling interaction [5].

An independent ISAC (also known as mono-static ISAC) system transmits and receives the signal by the same node simultaneously, and performs the SF on its own [6]. This approach requires full duplex transceivers with powerful self-interference suppression, but currently the suppression capability does not meet the performance required in a cellular network [7], [8]. To overcome this challenge, and to utilize the almost omnipresent base stations (BSs) and nodes in a cellular network, the cooperative ISAC (also known as bi-static and multistatic ISAC) system has been proposed where the transmitting and receiving nodes are separated and work at different states, that is, the transmitting node works at the downlink state while the receiving node works at the uplink state. Particularly for the BS node, the user equipment (UE) covered by the receiving BS can be muted or transmit uplink signal to the receiving BS [9], [10],

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[11]. Such cooperative ISAC can be realized with simply software upgrade to add SF in the existing mobile network. Large-scale communication nodes of various types can be utilized as sensing receivers, including BSs and UE [12], [13]. Sensing results of multiple receiving nodes can be fused to further enhance the sensing accuracy. Therefore, cooperative ISAC is widely favored by operators from the network perspective.

However, there are challenges in a cooperative ISAC system both at the link level and in network perspective. Synchronization among different sensing nodes is critical to ensure high sensing accuracy [14], while proper beam management to minimize searching overhead is of great importance [4], [15]. Sensing node placement for best sensing coverage [16], [17], number, and specific group of node selection as sensing receivers for optimal sensing accuracy, network interference management are all key issues to address in ISAC deployment by mobile operators [18]. Moreover, novel multiple-input multiple-output (MIMO) technologies, including pseudo MIMO and compact MIMO, may prove valuable in cooperative ISAC networks [19], [20]. These MIMO technologies not only improve spectral efficiency and energy efficiency of communication systems [21], but also help to miniaturize sensing nodes and enhance sensing accuracy.

In this article, we provide solutions to overcome the aforementioned challenges and explore potential advantages of the above MIMO technologies in the cooperative ISAC system. Based on the channel model of cooperative sensing, we address the key issues of ISAC system including synchronization, beam management and sensing algorithm. Then we study node placement and network interference management issues in an ISAC network. Further, we present consideration on the pseudo- and compact-MIMO techniques in cooperative ISAC system, showing their potential in ISAC system improvement. Finally, prototype designs at both sub-7GHz and millimeter wave bands are provided, and experiments are conducted in an outdoor environment to verify the performance.

This article is organized as follows. In "Sensing Model and Key Technologies" section introduces the sensing channel model of cooperative ISAC system and key technologies. In "Network ISAC" section describes the node placement and selection issues in ISAC network. "ISAC Enhanced by Novel MIMO" section introduces the novel MIMO techniques and its potential applications in ISAC systems. In "Prototype Design and Experimental Results" section provides prototypes of various cooperative ISAC systems and corresponding experimental results. In "Conclusion" section concludes with an outlook of future work.

Notation: The following notations are used throughout the article: Boldface lowercase and uppercase letters denote vectors and matrices, respectively. We denote the transpose, conjugate, and Hermitian as $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$, respectively. The notation $\mathbb{C}^M$ and $\mathbb{C}^{Q\times M}$ are the sets of M-dimensional vectors and $Q \times M$ matrices of complex numbers, respectively.

SENSING MODEL AND KEY TECHNOLOGIES

In this section, we first introduce the signal model of cooperative sensing. Then, to deal with the synchronization issue, which is a prerequisite of cooperative sensing, we introduce a synchronization scheme and a basic target sensing algorithm. Next, to minimize the scanning overhead with a large number of high-density narrow beams, we introduce a two-step beam management method for target sensing. Furthermore, to obtain the target imaging, we propose an artificial intelligence (AI) enabled point cloud sensing algorithm.

Sensing Model

3GPP established a new study item for ISAC channel modeling in December 2023. To model the multipath introduced by the target, the ISAC channel is separated into the target channel $h_{\mathrm{T}}(\tau,t)$ and background channels $h_{\mathrm{B}}(\tau,t)$ [22] as

$$h_{\mathrm{ISAC}}(\tau,t) = h_{\mathrm{T}}(\tau,t) + h_{\mathrm{B}}(\tau,t). \tag{1}$$

All target-related multipath components are grouped in the target channel, while the rest are in the background channel. Various propagation models between the transmitter (TX) and the receiver (RX) have been established in current standards, such as TR 36.777 [23], TR36.885 [24], and TR 38.901 [25]. These models can be used directly in the background channel here. Therefore, the key to modeling ISAC models is to characterize the target channel.

From the definition of radar cross-section (RCS), we know that the target intercepts the transmitted power from TX and then radiates the interception power in all directions. In this way, the target can be seen as a receiver in the TX-target channel and as a transmitter in the target-RX channel. The target channel is composed of the convolution between the TX-target link and the target-RX link

$$h_{\mathrm{T}}(\tau,t) = 10^{-\frac{PL_{\mathrm{tx}}+PL_{\mathrm{rx}}+SF_{\mathrm{tx}}+SF_{\mathrm{rx}}}{20}} \cdot \frac{2\sqrt{\pi}}{\lambda}\sqrt{\sigma_{\mathrm{RCS}}}\tilde{h}(\tau,t) \tag{2}$$

where PL_{tx} is the path loss from TX to the target, and SF_{tx} represents the corresponding shadow fading. Similarly, PL_{rx} and SF_{rx} denote the path loss and shadow fading from the target to the RX. Equation (2) is derived from the radar equation. It attempts to split the radar equation into TX-target and RX-target segments path loss model, which helps the reuse of the modeling results for each segment in the standard. The parameter σ_{RCS} is the large-scale RCS, a newly defined parameter in the 3GPP discussion. It is only related to the target type or size. $\tilde{h}(\tau,t)$ is further divided into the delay part and its coefficient part as

$$\tilde{h}(\tau,t) = \sum_{l=1}^{L}\tilde{h}_l(t)\delta\left(\tau - \tau_{\mathrm{tx},l} - \tau_{\mathrm{rx},l} - \frac{d_{\mathrm{tx,3D}}}{c} - \frac{d_{\mathrm{rx,3D}}}{c}\right) \tag{3}$$

where $\tau_{\mathrm{tx},l}$ is the delay relative to the line-of-sight (LOS) ray in TX-target channel. The propagation delay of TX-target LOS ray is $d_{\mathrm{tx,3D}}/c$. In conventional communication channels, it is usually only necessary to model the relative delay, i.e., the difference between the multipath delay and the delay of the direct path. This is because it is often assumed that the transceivers are already synchronized when evaluating performance. However,

the absolute propagation delay must be modeled in ISAC evaluation.

The detailed expression of $\tilde{h}_l(t)$ is as

$$\tilde{h}_l(t) = \sqrt{P_{\text{tx},l}P_{rx,l}}\mathbf{a}^{\mathbf{T}}(\theta_{\text{rx},l},\phi_{\text{rx},l})\mathbf{CPM}_{\text{rx},l}\mathbf{CPM}_{\text{target}} \cdot \mathbf{CPM}_{\text{tx},l}\mathbf{a}(\theta_{\text{tx},l},\phi_{\text{tx},l})\alpha_{\text{CPM}}e^{j2\pi\int_{t_0}^{t} f_{\text{D},l}(\tilde{t})d\tilde{t}} \quad (4)$$

where $P_{\text{tx},l}$ and $P_{\text{rx},l}$ are the ray powers in TX-target and RX-target channel, θ and ϕ denotes the azimuth and elevation angles. $\mathbf{a}(\theta_{\text{rx},l},\phi_{\text{rx},l})$ is the RX antenna pattern at azimuth angle $\phi_{\text{rx},l}$ and elevation angle $\theta_{\text{rx},l}$. It is a vector with two elements representing the antenna gain at vertical and horizontal polarizations, respectively,

$$\mathbf{a}(\theta_{\text{rx},l},\phi_{\text{rx},l}) = \begin{bmatrix} F_{\text{rx},\theta}(\theta_{\text{rx},l},\phi_{\text{rx},l}) \\ F_{\text{rx},\phi}(\theta_{\text{rx},l},\phi_{\text{rx},l}) \end{bmatrix}. \quad (5)$$

The parameter $\mathbf{a}(\theta_{\text{tx},l},\phi_{\text{tx},l})$ has the same expression by replacing the subscript. The Doppler $f_{\text{D},l}(\tilde{t})$ is a function of time $\tilde{t}$. This is to satisfy the need to monitor channel variations over long periods in sensing applications. The integration can be simplified to $f_{\text{D},l}t$ under instantaneous conditions. The cross-polarization matrices (CPMs) of TX-target and target-RX are $\mathbf{CPM}_{\text{tx},l}$ and $\mathbf{CPM}_{\text{rx},l}$. Their expression depends on the ray's LOS condition. For example, if the lth ray in the TX-target channel is LOS, its CPM is as

$$\mathbf{CPM}_{\text{tx},l} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (6)$$

If it is a none-LOS (NLOS) ray, its CPM expression is as

$$\mathbf{CPM}_{\text{tx},l} = \begin{bmatrix} \exp(j\Phi_l^{\theta\theta}) & \sqrt{\kappa_l^{-1}}\exp(j\Phi_l^{\theta\phi}) \\ \sqrt{\kappa_l^{-1}}\exp(j\Phi_l^{\phi\theta}) & \exp(j\Phi_l^{\phi\phi}) \end{bmatrix}. \quad (7)$$

The parameter κ_l in the equation is the cross polarization power ratio, and it follows Log-normal distribution. The initial phase $\{\Phi_l^{\theta\theta}, \Phi_l^{\theta\phi}, \Phi_l^{\phi\theta}, \Phi_l^{\phi\phi}\}$ are for four different polarization combinations $(\theta\theta, \theta\phi, \phi\theta, \phi\phi)$. They are drawn from a uniform distribution within $(-\pi, \pi)$. CPM of the target has the same expression with (7) by replacing the subscript. The parameter α_{CPM} is the normalization factor of CPM. It is used to ensure that multiplying three CPMs together does not result in an extra gain.

For the convenience of analyzing OFDM systems under instantaneous conditions, we substitute (4) into (3) and perform a Fourier transform on the time domain

$$\tilde{h}(f,t) = \sum_{l=1}^{L}\beta_l\exp(-j2\pi f\tau_l)\exp(j2\pi f_{\text{D},l}t) \quad (8)$$

where

$$\tau_l = \tau_{\text{tx},l} + \tau_{\text{rx},l} + \frac{d_{\text{tx,3D}}}{c} + \frac{d_{\text{rx,3D}}}{c} \quad (9)$$

$$\beta_l = \sqrt{P_{\text{tx},l}P_{\text{rx},l}}\mathbf{a}^{\mathbf{T}}(\theta_{\text{rx},l},\phi_{\text{rx},l})\mathbf{CPM}_{\text{rx},l}\mathbf{CPM}_{\text{target}} \cdot \mathbf{CPM}_{\text{tx},l}\mathbf{a}(\theta_{\text{tx},l},\phi_{\text{tx},l})\alpha_{\text{CPM}}. \quad (10)$$

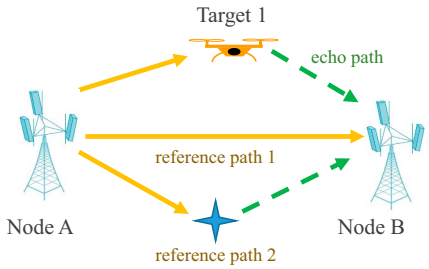


FIG. 1. Schematic diagram of time-frequency synchronization scheme.

Assuming the channel is measured at $t = n_sT_s$, $f = f_0 + n_c\Delta f$, the discrete channel can be expressed as

$$H(n_c,n_s) = \sum_{l=1}^{L}\beta_l e^{-j2\pi(f_0+n_c\Delta f)\tau_l}e^{j2\pi f_{\text{D},l}n_sT_s}. \quad (11)$$

Synchronization

In a practical sensing system, whether the time-frequency synchronization is well achieved directly impacts the sensing results accuracy. It is difficult to achieve ideal time synchronization between the clock signals of the transmitter and the receiver, thus clock deviations as well as Doppler frequency errors exist.

The reference path synchronization method is an important method used for time-frequency synchronization [26], where the transmitting node sends a known reference signal through the reference path, and the receiving node analyzes the received reference path signal to estimate the delay of the signal and calibrate the corresponding time-delay difference, thus achieving accurate time synchronization. Similarly, the receiver uses a phase-locked loop or a frequency shift estimation algorithm to estimate the Doppler frequency of the signal and calibrate the corresponding frequency difference. The schematic of the reference path for time-frequency synchronization is shown in Fig. 1. The proposed reference-path synchronization can be used in both LOS and NLOS environments. Although the received power of NLOS reference paths (e.g., reference path 2 in Fig. 1) is weaker than that of LOS paths, the known geometrical position of the reference paths allow efficient synchronization.

This synchronization method includes the following steps. First, the range and velocity estimation are conducted to obtain time delay and Doppler frequency for the reference path. Next, we calibrate the signal propagation range and velocity based on the estimated parameters and locations of nodes.

For a given receiving antenna of any receiver, we have the received signal frequency domain as $H(n_c,n_s)$ and write the received signal in matrix form $\mathbf{H}$, which is expressed as

$$H(n_c,n_s) = \sum_{l=1}^{L}\beta_l e^{-j2\pi(f_0+n_c\Delta f)\left(\frac{d_l}{c}+\tau\right)} e^{j2\pi T_s\left(\frac{f_c v_{\|,l}}{c}+f_D\right)n_s} + w(n_c,n_s) \quad (12)$$

where τ and f_D denote time delay error and Doppler frequency error of the transmitter concerning the n_rth receiver,

respectively, $\mathbf{H} \in \mathbb{C}^{N_c \times N_s}$ is a function of subcarrier index and symbol index. Next, we perform a two-dimensional discrete Fourier transform (DFT) to estimate the range and velocity

$$\tilde{\mathbf{H}} = \mathbf{D}_C \mathbf{H} \mathbf{D}_S^T \tag{13}$$

where $\mathbf{D}_C \in \mathbb{C}^{N_c \times N_c}$ and $\mathbf{D}_S \in \mathbb{C}^{N_s \times N_s L}$ are the Vandermonde matrix. The vector consisting of the diagonal elements of the Vandermonde matrix is the DFT vector, denoted as $[\mathbf{D}_c]_{n_c,n_c} = (1/N_c)e^{j(n_c-1)(2\pi/N_c)n_c}$. Then, we perform target detection to find the index $\hat{p}_l$ of the row and the index $\hat{q}_l$ of the column corresponding to the peak of the matrix $\tilde{\mathbf{H}}$. The propagation range d and the velocity v with targets are then solved as

$$\hat{d}_l = \frac{(\hat{p}_l - 1)c}{N_c \Delta f}, \quad \hat{v}_l = \frac{(\hat{q}_l - 1)c}{N_s T_s f_c}. \tag{14}$$

The values of the corresponding maxima indicate the complex amplitude $\tilde{\beta}_l$ of the reference path. The nearest path is generally considered to be the direct path (reference path), the time synchronization error between nodes is obtained based on the difference between the range of the estimated reference path and the range between nodes, and the frequency synchronization error between nodes is obtained based on the velocity of the estimated reference path. In later target estimations, we can subtract the time synchronization error and the frequency synchronization error from the estimated target time delay and Doppler, respectively, to obtain the final estimation results. The method supports dynamic scenarios with moving targets where BS, building, terrain, etc., remain static.

Round-trip time (RTT) measurement enables synchronization through two-way signal propagation, though the time and efficiency cost of additional measurement are doubled [27]. Fiber-based synchronization using physical connections achieves high accuracy but suffers from limited deployment flexibility. In contrast, our reference-path method leverages existing geometric relationships between nodes, requiring no additional hardware cost and round-trip measurements.

Beam Management

Beam scanning without prior knowledge of target directions is required in practical scenarios. High directivity of a beamformer implies narrow spatial coverage per beam, and a large number of beams for full spatial coverage. Each beam requires dedicated reference signal resources and measurement reporting [e.g., reference signal received power (RSRP)/signal to interference plus noise ratio (SINR)], leading to significant resource and signaling overhead. While insufficient beam density risks missing target alignment, resulting in weak reflected signals and degraded sensing accuracy. This tradeoff motivates our pursuit for optimized beam configuration strategy.

To balance overhead and performance, we adopt a two-phase beam management scheme with beam oversampling. This scheme determines the coarse target range in the first phase and the fine beam direction in the second phase.

1) *Phase 1—Coarse Detection*
   a) Node A transmits signals through different analog beams.
   b) Node B measures RSRP/SINR across beams and reports to Node A.
   c) Node A identifies adjacent beams satisfying

$$|\mathrm{RSRP}_i - \mathrm{RSRP}_j| < T \text{ and } \arg\max_k(\mathrm{RSRP}_k) \in \{i, j\}, \tag{15}$$

   where T is a preset threshold, and i, j denote adjacent beam indices. If the condition is satisfied, go to Phase 2. Otherwise, Node A chooses the beam with largest RSRP/SINR value reported by Node B.
2) *Phase 2—Refined Beamforming*
   a) Node A generates oversampled beams between qualified adjacent beams.
   b) Node B evaluates and reports optimal oversampled beam indices.

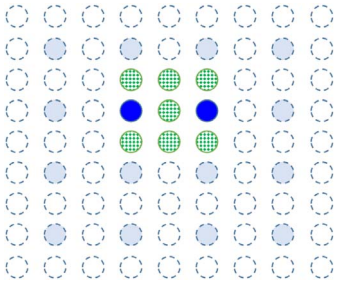


FIG. 2. Beam oversampling configuration between two adjacent beams (blue: original beams, green: oversampled beams).

An important aspect of the beam management is the generation of oversampled beams. The oversampled beam generation process is described below. Assume that the two strongest beams are horizontally adjacent, as shown in the blue circles in Fig. 2, corresponding to beam indices (k, l) and $(k, l+1)$. For beam (k, l), the vertical and horizontal phases of the antenna element are $\phi_{v,k} = 2\pi k \cdot m/M$ and $\phi_{h,l} = 2\pi l \cdot n/N$, respectively. Similarly, for beam $(k, l+1)$, the vertical and horizontal phases of the antenna element are $\phi_{v,k} = 2\pi k \cdot m/M$ and $\phi_{h,l+1} = 2\pi(l+1) \cdot n/N$, respectively.

Here, the beam direction is refined by increasing the oversampling factors O_v and O_h. For example, if both the vertical and horizontal directions are refined into two times the beams, then the corresponding oversampling factors O_v and O_h are both 2. In this case, the original beam (k, l) corresponds to vertical and horizontal phases of $2\phi_{v,k}/O_v$ and $2\phi_{h,l}/O_h$, respectively, and the original beam $(k, l+1)$ corresponds to vertical and horizontal phases of $2\phi_{v,k}/O_v$ and $2\phi_{h,l+1}/O_h$, respectively. The vertical and horizontal phases of the remaining seven beams (green circles) are $2\pi a \cdot m/(m \cdot O_v)$ and $2\pi b \cdot n/(N \cdot O_h)$, respectively, where the values of the (a, b) combination are $(2k-1, 2l), (2k-1, 2l+1), (2k-1, 2l+2), (2k, 2l+1), (2k+1, 2l), (2k+1, 2l+1)$, and $(2k+1, 2l+2)$. The generated vertical and horizontal precoding matrices are as follows:

$$\mathbf{W}_v = \left[1, e^{j2\pi a/(M \cdot O_v)}, \cdots, e^{j2\pi a(M-1)/(M \cdot O_v)}\right] \tag{16}$$

$$\mathbf{W}_h = \left[1, e^{j2\pi b/(N \cdot O_h)}, \cdots, e^{j2\pi b(N-1)/(N \cdot O_h)}\right]. \tag{17}$$

And the final synthesized precoding matrix is as follows:

$$\mathbf{W} = \left[\mathbf{W}_h, e^{j2\pi a/(M\cdot O_v)}\mathbf{W}_h, \cdots, \mathbf{W}_h e^{j2\pi a(M-1)/(M\cdot O_v)}\right]. \quad (18)$$

Multitarget sensing can be implemented by time division or multipanel (each panel generates one beam). In addition, if two targets are near to each other and served by one analog beam, different digital precoder with finer beam can also be generated under hybrid beamforming structure to divide different targets in space dimension.

AI Sensing Algorithm

The previous subsections have examined sensing algorithms and beam management strategies under LOS or sparse multipath conditions. However, for realistic and complex targets, especially in 3-D scenarios, multiple scattering centers often exist. These scattering interactions render the electromagnetic (EM) inverse problem highly nonlinear and severely ill-posed, making traditional AI models ineffective in capturing the full spatial variation of EM properties [28].

Conventional AI models, such as convolutional neural networks (CNNs) [29], recurrent neural networks (RNNs), and standard transformer-based architectures [30], often suffer from limited generalizability when applied to high-dimensional inverse scattering problems. Even with large training datasets, these models are typically trained as black-box regressors without explicitly incorporating the underlying physical principles of EM propagation [31], [32], [33], [34]. Consequently, they exhibit high sensitivity to noise, poor robustness to unseen geometries, and excessive computational cost during inference [35], [36].

In contrast, model-assisted or physics-informed neural networks (PINNs) [35] have emerged to integrate physical constraints into neural architectures. Although these methods show promise, they often struggle to capture multimodal posterior distributions inherent in inverse EM problems, especially under multiple scattering conditions.

To overcome these limitations, we propose the use of diffusion models, which are generative models designed to learn the posterior distribution of complex data through a sequential denoising process [37]. Recent studies [38], [39] have demonstrated that diffusion models are inherently more robust to measurement noise, better capture uncertainty in inverse predictions, and offer competitive performance even in low-data regimes. These properties are especially advantageous for EM property sensing, where the inverse mapping from backscattered field observations to permittivity profiles is ill-conditioned and sensitive to perturbations.

The key motivations for choosing diffusion models in this work are summarized as follows. *1) Robustness to Noise*: Diffusion models perform iterative denoising and thus provide stable reconstructions under low SNR. *2) Probabilistic Inference*: Unlike deterministic AI models, diffusion models sample from a learned distribution, yielding more reliable results with uncertainty quantification. *3) Generalization Across Scenarios*: Trained diffusion models can generalize better to unseen target configurations and measurement domains.

Therefore, we adopt a conditional diffusion model framework to solve the 3-D EM property sensing task within an ISACs system. Our method jointly utilizes communication and sensing reflections to construct the conditional input, enabling accurate and robust EM property estimation. We first utilize both communications and sensing signals that are reflected back from the target to estimate the sensing channel. Suppose the lth transmit signal at the BS is expressed as

$$\begin{aligned}\mathbf{x}(l) &= \mathbf{W}_\mathrm{s}\mathbf{s}(l) + \mathbf{W}_\mathrm{c}\mathbf{c}(l)\\ &= [\mathbf{W}_\mathrm{s}, \mathbf{W}_\mathrm{c}]\left[\mathbf{s}(l)^\top, \mathbf{c}(l)^\top\right]^\top\\ &= \mathbf{W}\hat{\mathbf{x}}(l)\end{aligned} \quad (19)$$

where $\mathbf{s}(l) \subseteq C^{N_t\times 1}$ denotes the dedicated sensing signal and $\mathbf{c}(l) \subseteq C^{K\times 1}$ denotes the transmission symbols intended to K UEs. Moreover, $\mathbf{W}_\mathbf{s} = [\mathbf{w}_{s,1}, \mathbf{w}_{s,2}, \cdots, \mathbf{w}_{s,N_t}] \subseteq C^{N_t\times N_t}$ and $\mathbf{W}_\mathrm{c} = [\mathbf{w}_{c,1}, \mathbf{w}_{c,2}, \cdots, \mathbf{w}_{c,k}] \subseteq C^{N_t\times K}$ represent the digital beamforming matrices for sensing and communications, respectively. Besides, $\mathbf{W} = [\mathbf{W}_\mathrm{s}, \mathbf{W}_\mathrm{c}]$ denotes the equivalent transmit beamforming matrix. We then adopt the least squares to estimate channel $\tilde{\mathbf{h}}_s$, i.e.,

$$\widehat{\tilde{\mathbf{h}}}_s = (\mathbf{X}^*\mathbf{X}^\top \otimes \mathbf{I}_{N_r})^{-1}(\mathbf{X}^\top \otimes \mathbf{I}_{N_r})^H\tilde{\mathbf{y}} \quad (20)$$

where $\mathbf{X} = [x(1), \cdots, x(L)]$ denote the transmitted signal over L symbols. I_{N_r} denote the identity matrix with compatible dimensions; the received target echo signals at the BS is expressed as Y. $\hat{y} = \mathrm{vec}(\mathbf{Y}) \in \mathbb{C}^{N_r L\times 1}$ The echo electric field at the BS's receiver scattered back from the target can then be formulated as

$$\mathbf{E}^s(\mathbf{r}_n, l) = k_b^2 \int\int\int_D \overline{\overline{\mathrm{G}}}(\mathbf{r}_n, \mathbf{r}')\chi(\mathbf{r}')\mathbf{E}^t(\mathbf{r}', l)\mathrm{d}\mathbf{r}' \quad (21)$$

Suppose the target is located within the region D, where $\mathbf{r}_n$ denotes the position of the nth receiving antenna. $\overline{\overline{\mathrm{G}}}(\mathbf{r}_n, \mathbf{r}') \in \mathbb{C}^{3\times 3}$ is the dyadic electric field Green's function. k_b is the wave number in the background medium. The contrast function $\chi(\mathbf{r}')$ to be reconstructed defined as the difference in complex relative permittivity between the target and the background medium (the air here). $\mathbf{E}^t$ is the total electric field within D.

Suppose the receiver can only measure the scalar electric field component in the direction represented by the unit vector $\hat{\mathbf{p}} \in \mathbb{R}^{3\times 1}$. The lth received echo signals can be formulated as

$$\mathbf{y}(l) = G_r[\mathbf{E}^s(\mathbf{r}_1, l), \cdots, \mathbf{E}^s(\mathbf{r}_{N_r}, l)]^\top\hat{\mathbf{p}} + \mathbf{z}(l) \quad (22)$$

where G_r is the receiving antenna gain. According to (20) and (22), we can get the estimate channel $\tilde{\mathbf{h}}_s$ for point cloud reconstruction.

Let $\mathbf{x}_i^{(1:T)}$ denote the set of $\mathbf{x}_i^{(1)}, \ldots, \mathbf{x}_i^{(T)}$ that are sequentially generated from $\mathbf{x}_i^{(0)}$ with the maximum time step T. Once the sensing channel is estimated, we employ the diffusion model to generate a detailed point cloud representation of the target. The forward diffusion process can be deliberately modeled as a Markov chain, i.e.,

$$q\left(\mathbf{x}_i^{(1:T)}|\mathbf{x}_i^{(0)}\right) = \prod_{t=1}^{T} q\left(\mathbf{x}_i^{(t)}|\mathbf{x}_i^{(t-1)}\right). \quad (23)$$

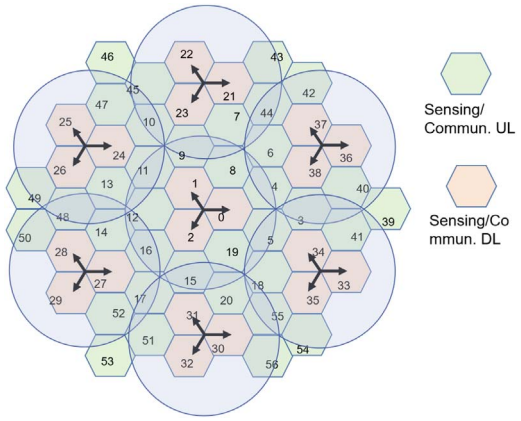


FIG. 3. Ring networking for cooperative ISAC.

Similar to the forward diffusion process, the reverse diffusion process can also be modeled as a Markov chain conditioned on the estimated sensing channel, i.e.,

$$p_{\boldsymbol{\theta}}\left(\mathcal{X}^{(0:T)}|\hat{\mathbf{H}}_s\right) = \prod_{i=1}^{N} p_{\boldsymbol{\theta}}\left(\mathbf{x}_i^{(0:T)}|\hat{\mathbf{H}}_s\right). \tag{24}$$

Define the ith normalized nondimensional 5-D point $\mathbf{x}_i^0 \in \mathbb{R}^{5\times 1}$ that comprises both the 3-D location information and the 2-D EM property as

$$\mathbf{x}_i^{(0)} = \left[\frac{x_i - x_c}{x_d}, \frac{y_i - y_c}{y_d}, \frac{z_i - z_c}{z_d}, \frac{\epsilon_i}{\epsilon_0}, \frac{\sigma_i}{\epsilon_0 \omega}\right]^{\top}. \tag{25}$$

This point cloud provides a clear 3-D visualization of the EM property distribution, including critical parameters such as shape, relative permittivity, and conductivity.

NETWORK ISAC

In this section, we study the cooperative ISAC system from a network perspective. Starting with the interference issue and potential resolution in the ISAC network then move on to the sensing node placement and selection for cooperative sensing.

Network Interference Management

Comparing to communication networks, interference issue in ISAC network is much more severe so that network interference management for ISAC systems required investigation.

As shown in Fig. 3, we consider cooperative ISAC in a cellular network, where each downlink site combined with three sectors are surrounded by six uplink sites. In each downlink sector, the BS can be a sensing or communication transmitter. When the BS serves as the sensing transmitter, it transmits sensing signal to a sensing receiver in the neighboring uplink sector. While when it serves as the communication transmitter, it transmits the signal to its serving communication UEs. The BS in other uplink sectors are implemented to receive sensing signals or its' own serving communication UEs. We name such networking scheme as "Ring" ISAC networking. To enhance the received sensing power, both the transmitters and the receivers employ beamforming, directing their beams towards the sensing target. In particular, the beamforming management scheme proposed in the "Synchronization" section can be applied to reduce the system overhead. For a particular sensing receiver, the interference comes from power leakage from the sensing transmitters located in different sectors. The intensity of the sensing interference is relevant to the central frequency, the relative positions between the sensing received sector and the interfering sectors, and the beam directions. We consider UAV sensing at the cellular network with 19 sites 57 sectors in both 2.6 and 26 GHz system. The inter-site distance (ISD) is set as 100, 200, and 300 m. The sensing receiver in sector 4 receives both the sensing transmitted signal from sector 1 and the interference from other downlink interference BSs. The sensing targets are randomly placed in the coverage area. Table I presents the interference level and the corresponding received sensing SINR of a particular sensing receiver at 90 percentile. Ring networking without particular interference management are applied. As seen, without network interference management, the received sensing SINR is fairly weak, especially for the long-range sensing, i.e., ISD larger than 300 m.

TABLE I. Interference and sensing SINR of UAV over different carrier frequencies

Frequency (GHz)	26			2.6		
ISD (m)	100	200	300	100	300	500
Interference (dBm)	−47.33	−53.55	−56.60	−24.50	−28.33	−33.22
SINR of UAV (dB)	−2.69	−3.05	−14.86	−13.45	−30.62	−35.49

Based on the distribution of sensing interference, the interfering sectors are classified into three levels.

1) *The First-Level Interfering Sectors*: It refer to those in the same site as the transmitted sector and those in neighboring sites whose transmitted signal emitted by the BS pointing towards the sensing received sector.
2) *The Second-Level Interfering Sectors*: It refer to the sectors at a range less than 2 ISD w.r.t the sensing received sector.
3) *The Third-Level Interfering Sectors*: It refer to the sectors at a range more than or equal to 2 ISD w.r.t the sensing received sector.

For instance, in Fig. 3, consider sectors 0 and 4 as the sensing transmitted and sensing received sector, respectively, the first-level interfering sectors are sector 1, 2, 34, 35, 37, and 38. The second-level interfering sectors are sector 33 and 36. The third-level interfering sectors are sector 21–30.

Accordingly, two stages of the network interference managements are applied, which respectively eliminate the first and the second levels of interfering sectors and keep the third-level interfering sectors as residual interfering sectors. One approach is to assign orthogonal codes to the transmitted sector, the first-level interfering sectors, and the second-level interfering sectors, thereby reducing the interference to a

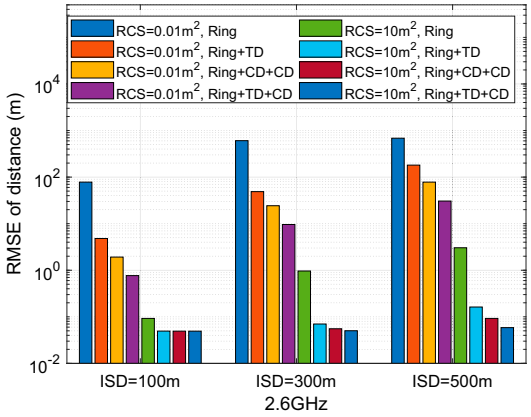


FIG. 4. Comparison of the RMSE of the range estimation over three interference management methods at 2.6 GHz.

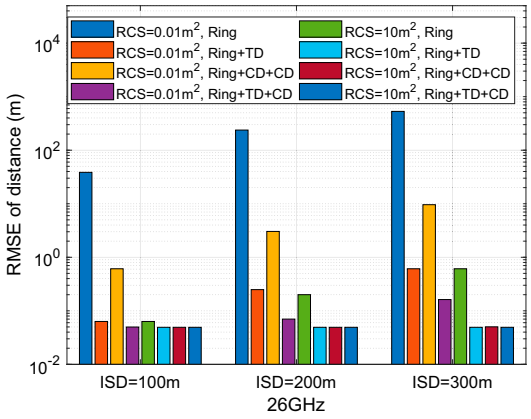


FIG. 5. Comparison of the RMSE of the range estimation over three interference management methods at 26 GHz.

relatively low level. We name such interference management method as "Ring+CD+CD." Alternatively, time domain interference management method is proposed. In the first stage of such interference management, named "Ring+TD," time division scheduling among three sectors of the same site can be applied to cancel the first-level interference. In the second stage of such interference management, named "Ring+TD+CD," code orthogonality among second-level interfering sectors are applied on the basis of "Ring+TD."

Using the *Ring* ISAC networking as the baseline in the evaluation, Figs. 4 and 5 compare the RMSE of range sensing under various network interference management methods described above. The performance at 2.6 and 26 GHz for two types of targets, UAV with RCS of 0.01 m^2 and Car with RCS of 10 m^2 are presented. Generally, the network interference in 2.6 GHz is higher than that in 26 GHz, while the RMSE of range in UAV sensing is larger than that in car sensing (lower than 0.1 m).

In time domain interference management methods, triple resource overhead is introduced, and the sensing performance is improved more than ten times. For instance, in 2.6 GHz UAV sensing, the RMSE of range decreases from 78 m under "Ring" to 5 m under "Ring+TD" when ISD equals to 100 m. Such impressive improvement is even more significant in 26 GHz system. Therefore, "Ring+TD" and "Ring+TD+CD" appear to be very effective.

In the case of Car sensing, the RMSE of range improvements under various interference management methods are quite similar, all providing apparent RMSE performance gain. "Ring+CD+CD" would be preferred for the time domain resource efficiency. However, in the cases of UAV sensing, "Ring+CD+CD" is apparently worse than time domain methods at 26 GHz. That's because the interference is severe in the case of UAV sensing at 26 GHz, and the residual interference under "Ring+CD+CD" still leads to a significant deterioration in range estimation accuracy.

Therefore, despite consuming additional time resource, time domain interference management method, like "Ring+TD" and "Ring+TD+CD", is essential for the case of UAV sensing at 26 GHz. Whereas for all other cases, including UAV sensing at 2.6 GHz and Car sensing at both bands, it is sufficient to apply "Ring+CD+CD" method for interference management.

The interference management can be further improved by including the beam management method in the "Synchronization" section in network ISAC systems. From the perspective of transmitter sector in the Ring network, using the proposed beam management strategy can enhance the received SINR and sensing accuracy. Meanwhile, from the interfering sector's perspective, the proposed beam management can effectively reduce leakage interference. Last, applying the proposed low overhead beam management scheme can reduce the total time delay of the interference management procedure.

Node Placement and Selection

From the network perspective, a variety of sensing nodes may participate in the sensing service, including: 1) BSs with fixed positions such as macrosites, microsites, picosites and femtosites; 2) moving nodes including UAVs, intelligent robots; 3) UE of mobile phones and other handheld or wearable devices; and 4) sensing exclusive devices such as customized sensing terminals. Naturally, it is desirable to maximize the sensing accuracy with minimum number of sensing nodes.

These sensing nodes fall into two categories, namely static versus moving nodes. Static nodes are normally equipped with more powerful hardware, such as larger antenna array (larger aperture) and faster processing speed, compared with moving nodes. However, the density of static nodes is generally lower than moving nodes, hence a static node must cover a larger sensing area than a static counterpart. We investigate the placement of static nodes to maximize the minimum received sensing signal power within the sensing area. Considering cooperative ISAC between a macrosite and microsites within the microcell coverage area, the density of microsites close to the cell edge should be higher to compensate for the large path loss as shown in [40]. The selection process of static nodes can be trained offline and, given a certain sensing area, the optimal group of sensing nodes are fixed.

The moving nodes can change their positions when compared to the static nodes with fixed positions. However, the moving nodes should be static during the synchronization and

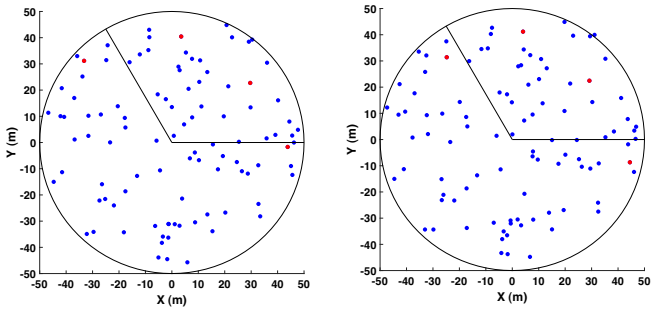


FIG. 6. Optimal node selection in two consecutive sensing period.

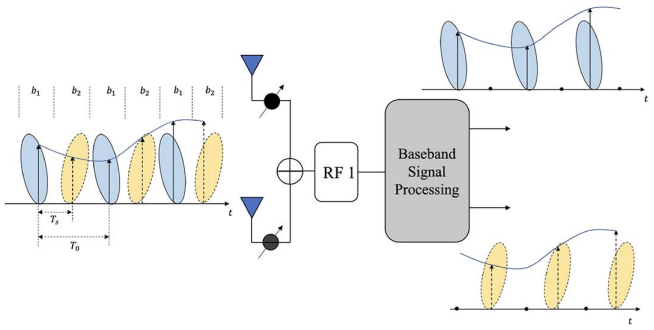


FIG. 7. Illustration of sample-level analog beamforming reconfiguration, upsampling, and signal splitting. T0 and Ts represent the original OFDM sampling period and the analog beamforming pattern switching period, respectively.

sensing process. However, both the node selection complexity and the selection update frequency are higher. The selection of moving sensing nodes requires in-time calculation and optimization. To start with, the selection of moving nodes is challenging since the nodes position keep varying. Those moving nodes satisfied speed limits can be potential sensing receivers. This is to mitigate the impact of movement on sensing signal propagation delay or Doppler frequency. Then the synchronization between the BS and moving nodes needs to be performed. The moving nodes need to be static before completing the synchronization process. The estimation of time and frequency synchronization error are then performed using the synchronization method in the "Synchronization" section. The synchronization results can be used for a following certain amount of time so that repeating the synchronization process are required. Afterwards, moving nodes position information should be reported to the sensing server for it to update optimal node selection. To reduce the computational complexity, a-priori information can be used. For example, the node selection can be optimized based on the optimal selection and their surrounding nodes in the last sensing period since the moving speed is low. The selected nodes will then receive the instruction and authorization from the sensing server to estimate the parameters of sensing objects. Fig. 6 shows optimal selection of moving nodes by minimizing Cramer-Rao Bound (CRB) [41], in two consecutive sensing periods with one second separation. A static sensing transmitter is located at the center of a circle with 50 m radius, and the area of sensing is a 120° sector of that circle. 100 moving nodes (blue points) are randomly distributed within the circle. The speed of moving nodes are below 2 meters per second with random directions. It can be observed that the optimal group of moving nodes (red points) has already changed for two consecutive seconds. Therefore, the selection of moving nodes is feasible yet much more challenging.

ISAC ENHANCED BY NOVEL MIMO

In this section, we explore two special MIMO designs and potential benefit they may bring to the cooperative ISAC systems, namely Pseudo MIMO and Compact MIMO.

Pseudo MIMO

Pseudo MIMO is an innovative MIMO-OFDM communication design that enables the transmission of more parallel data streams than available receiving RF chains [19]. As illustrated in Fig. 7, the core principle of pseudo MIMO involves configuring multiple distinct analog beamforming patterns for an antenna array connected to a single receiving RF chain and sequentially switching among these patterns within one original OFDM sampling period. By upsampling the received signal, the receiver captures the same OFDM symbol through different beamforming patterns, effectively supporting more parallel data streams than the available RF chains, as if additional nonphysical RF chains were equipped at the receiver. Unlike conventional hybrid MIMO systems, pseudo MIMO enhances spectral efficiency while maintaining comparable hardware complexity and power consumption. Moreover, compared to FD MIMO systems, it achieves equivalent spectral efficiency with substantially lower hardware costs and reduced power consumption.

Pseudo MIMO offers significant advantages for ISAC applications by improving sensing accuracy, a key performance metric. As noted in [3], sensing accuracy is directly proportional to the received SNR. The estimation error of the time delay as well as Doppler frequency are related to the received SNR by

$$\tilde{\tau} \geq \sqrt{\frac{3c^2}{8\gamma B^2}} \tag{26}$$

$$\tilde{f}_D \geq \sqrt{\frac{3\lambda^2}{8\gamma T^2}} \tag{27}$$

where $\tilde{\tau}$ and $\tilde{f}_D$ represent the estimation error of time delay, Doppler frequency, γ and λ denote received SNR and wavelength, B and T refer to the total bandwidth and sensing time, respectively.

In contrast to hybrid MIMO systems, pseudo MIMO increases the received SNR by enhancing receiving diversity, thereby improving sensing accuracy. A basic example can be given: A transmitter is equipped with a single RF chain connected to a single antenna element, and a receiver is equipped with a single RF chain connected to two antenna elements. Without pseudo MIMO technology, the receiving diversity of

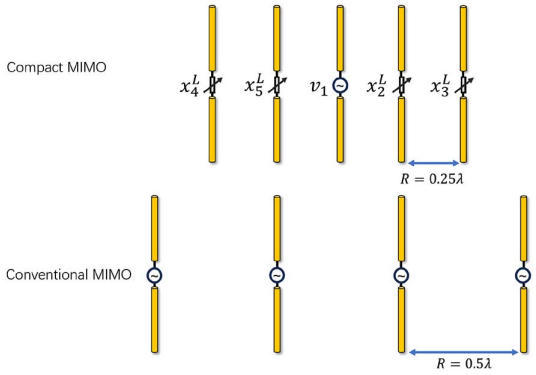


FIG. 8. Example of compact MIMO antenna with five dipole elements.

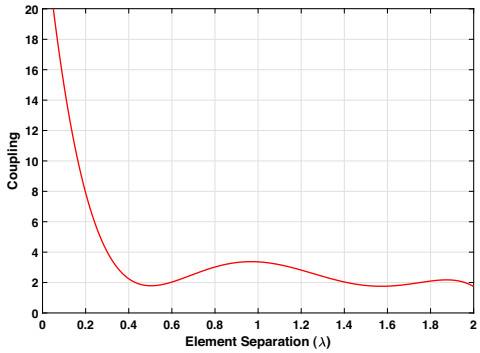


FIG. 9. Condition number of impedance matrix for five-element single-RF compact MIMO antenna.

the system would be one due to the single receiving RF chain. However, by implementing pseudo MIMO, the diversity increases to two. Under Rayleigh fading channels, enhanced diversity results in a 3 dB increase in the received SNR. This improvement in SNR reduces the estimation error, the metric for sensing accuracy, by 30%. Furthermore, many ISAC systems rely on an orthogonal resource allocation scheme where sensing and communication functions use separate resources. By outperforming hybrid MIMO in communication, pseudo MIMO frees up additional resources for sensing without compromising communication performance, ultimately enhancing the overall ISAC performance.

Compact MIMO

Compact MIMO antenna refers to the multiple antenna configuration with element separation significantly smaller than half wavelength, which greatly reduce the overall size of RF front-end at the expense of array aperture. Therefore, a single compact MIMO antenna is suitable for usage scenarios with limited size and form factors at the UE side. However, the antenna elements in compact MIMO suffers from severe mutual coupling effect so that the resulting radiation efficiency is reduced as the number of antenna element increases. To overcome this issue, an effective solution is reducing the number of RF chains, e.g., single-RF MIMO antenna [42]. This is often associated to antenna reconfigurability which provides additional dimension for imitating MIMO purpose. In single-RF MIMO as shown in Fig. 8, only one active element is connected to an RF chain with the other parasitic elements coupled to the active element. By tuning the variable loads of lumped elements at the parasitic elements in compact MIMO, the antenna radiation patterns are changed so that multiple streams of received signals can be utilized for sensing. The design of single-RF compact MIMO should optimize the tradeoff between the antenna element separation and robustness. To quantify mutual coupling effect, we can use the condition number of impedance matrix as shown in Fig. 9. When the element separation is very small, the condition number is very large due to strong mutual coupling. As a result, the compact MIMO is very sensitive to small error of load values, leading to weak robustness of the antenna system. In practice, condition number in the range of 5 to 10 is suitable for a good compact MIMO design.

By using the single-RF compact MIMO as the moving sensing nodes, single-RF compact MIMO antennas can improve the propagation delay and Doppler frequency estimation performance in ISAC system by improving SNR which is similar to pseudo MIMO as in (26) and (27). This is because reducing RF chain number improve the overall energy efficiency of MIMO antenna, implying that higher SNR is achieved with same amount of input power when compared to conventional MIMO. While for the angle estimation, smaller aperture of compact MIMO antenna results in worse accuracy according to

$$\tilde{\phi} \geq \sqrt{\frac{0.785 \times 3\lambda^2}{8\gamma D^2}} \tag{28}$$

where $\tilde{\phi}$ and D are the angle estimation error and antenna aperture. When the antenna separation is a quarter wavelength and SNR is improved by 3 dB, the angle estimation error is enlarged by 1.4 times [43]. However, it should be noted that the configuration of compact MIMO antenna is more flexible than the uniform antenna array, making it fit the requirement of the handheld devices.

To further expand the scale of compact MIMO antenna for BS in cooperative ISAC systems, we can implement multiple single-RF antenna in an conventional array of hybrid MIMO antenna [21], as shown in Fig. 10, where each single-RF antenna is connected with an RF chain with separation among single-RF being half wavelength [20]. In other words, each single-RF compact antenna take the place of antenna elements that are connected to the single RF chain in conventional MIMO. The advantages of this approach is that the analog devices of phase shifters can be replaced by tunable lumped elements. The overall cost of the RF front-end can be further reduced and the improvement in energy efficiency can be greater. In addition, the reconfigurability of compact MIMO antenna benefits the beamforming gain of antenna array, which is also desirable for cooperative ISAC system. It has been shown in [20] that this architecture achieves sensing performance close to the conventional MIMO antenna with 1 dB SNR while it improves energy efficiency by 4 dB,

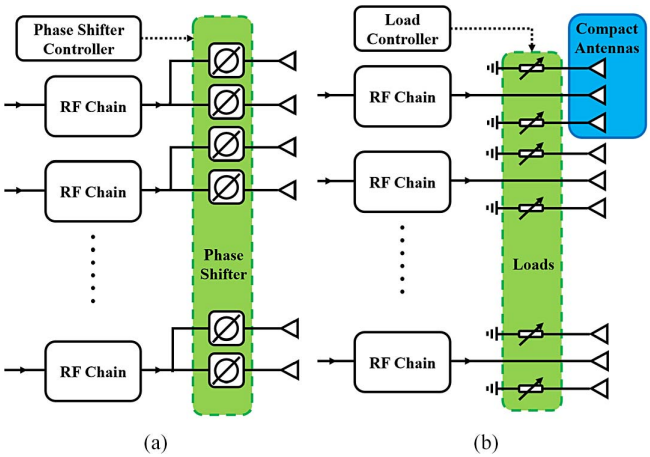


FIG. 10. Architectures of large-scale: (a) conventional hybrid MIMO; and (b) compact MIMO.

demonstrating the effectiveness of the architecture in [20] in cooperative sensing.

PROTOTYPE DESIGN AND EXPERIMENTAL RESULTS

In this section, we present three prototype designs that experimentally validate key theory of our ISAC framework. The millimeter-wave prototype demonstrates synchronization error mitigation method, achieving sub-meter accuracy through our reference-path synchronization method. The sub-7GHz prototype verifies multinode cooperative ISAC implementation using existing 5G infrastructure, while the pseudo MIMO prototype validates improved sensing accuracy compared with hybrid MIMO of comparable hardware cost.

Prototype at Millimeter Wave Band

Prototype Design and Parameters

The ISAC prototype at millimeter-wave band is designed to process received OFDM signals through a sequence of operations, including signal sampling, demodulation, and digital signal processing. High-speed analog-to-digital converters (ADCs) capture incoming millimeter-wave signals, ensuring accurate sampling of the received OFDM waveforms. The sampled signals undergo a fast Fourier transform (FFT) to extract frequency-domain subcarriers. Channel estimation and equalization techniques are applied for robust signal recovery, followed by digital signal processing (DSP) operations to estimate target parameters with high accuracy.

The center frequency of the prototype is 26 GHz. Both the sensing bandwidth and communication bandwidth are 200 MHz. The system is equipped with 256 antennas in a 16-by-16 array, and the equivalent isotropic radiated power (EIRP) is 58 dBm. Both the sensing and communication utilize the OFDM waveform. For the 6G frame structure, the sensing symbols can be configured anywhere in the time slot where the sensing and communication functions uses a time division scheme with a sensing resource allocation ratio of 1/28.

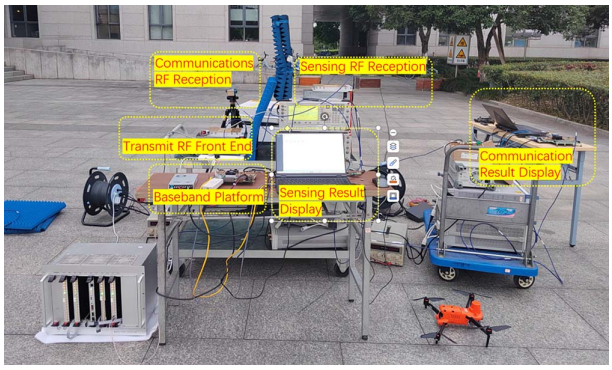


FIG. 11. Prototype of the cooperative sensing system.

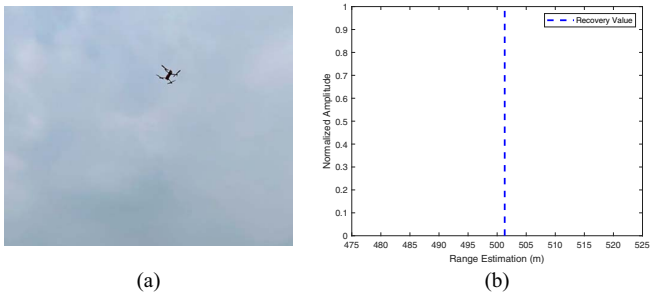


FIG. 12. (a) UAV scene. (b) Recovery result.

Sensing Accuracy Trial

In the experiments, the two BSs are synchronized using the fiber connection. The system-level testbed architecture of the cooperative sensing system is shown in Fig. 11. The hardware of the system is composed of RF front-end, baseband board which generates and processes the reflected signals. After passing signal processing, the range, angle and velocity information of the UAV are obtained.

In the field experiment, the propagation range of the path with UAV is around 500 m. The test scenario is shown in Fig. 12(a). The target locates right above the transmitter at 500 m altitude. An illustrative measurement result of the obtained range estimation is displayed in Fig. 12(b) where the estimated value of the UAV altitude is obtained. Through several trials, we obtain sub-meter range estimation accuracy, which demonstrates the effectiveness of the proposed prototype [11].

Synchronization Method Trial

The synchronization scheme verification setup is illustrated in Fig. 13. The prototype comprises a transmitter, receiver, UE, and processing unit. In the experiments, two BSs are synchronized using the reference path-based synchronization method in the "Synchronization" section and target sensing using the DFT processing in the "Synchronization" section. By transmitting the communication reference signal, the receiver captures echo signals reflected by the target and its processing unit applies technique in the "Synchronization" section to perform synchronization. UE receives communication signals and measures the transmission rate at the same time.

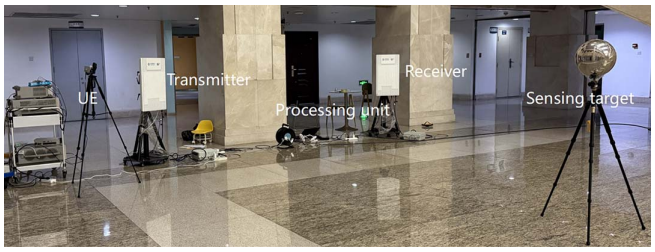


FIG. 13. ISAC prototype trial scenario diagram.

To evaluate the performance of synchronization techniques, experiments are performed with a test range of 4 to 6 m. Targets will be placed within this range to evaluate synchronization accuracy and range estimation performance. Synchronization between transmitter and receiver will be achieved using both fiber direct connection and air interface for calibration and evaluation. After synchronization, range of targets will be measured and the estimation errors will be analyzed. Experimental results show a range estimation accuracy of 0.2 m in the case of air interface synchronization and 0.16 m in the case of direct fiber connection.

The communication performance is also conducted. A horn antenna acts as UE to receive and process communication signals. The system utilizes 16-quadrature amplitude modulation (QAM) to transmit a data stream, achieving a transmission rate of approximately 683.78 Mbps without pilot signal overhead.

In addition, precommercial products based on this synchronization scheme for cooperative sensing have been developed. Trials at 4.9 GHz in multiple cities have demonstrated a positioning accuracy of 10 m with 1° angular estimation error.

Preliminary Prototype at Sub-7 GHz Band

Hardware and Deployment

In addition, a sub-7 GHz prototype is developed based on the existing 5G network infrastructure, aiming at leveraging multinode cooperative ISAC system. This prototype consists of multiple active antenna units (AAUs) operating in the sub-7 GHz band, with each AAU configured as a dual-polarized uniform planar array (UPA). The UPA elements are arranged in both azimuth and elevation dimensions, enabling flexible beamforming for improved sensing and communication performance. For the UPA configuration, the number of elements are equivalent to the number of RF chains in the azimuth dimension, while multiple antenna elements share a single RF chain in the elevation dimension. The deployment scenario includes multiple BSs acting as transmitting and receiving nodes, forming a cooperative sensing network. The block diagram of the sub-7 GHz prototype receiving node for sensing is shown in Fig. 14. This sub-7 GHz prototype mainly comprises an AAU, a building baseband unit (BBU), and a sensing function (SF) component. The AAU consists of a radio frequency (RF) unit and an FPGA, primarily used to receive RF signals and acquire ISAC information. It then transmits these data to the BBU via a common public radio interface (CPRI). The BBU includes an FPGA, Layer 1 (L1), and Layer 2 (L2). For

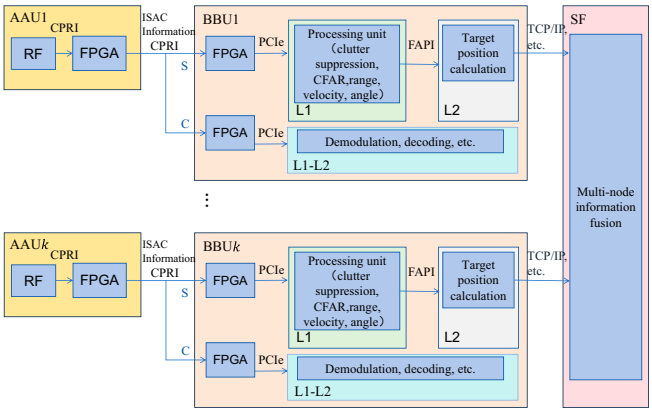


FIG. 14. Block diagram of sub-7 GHz prototype.

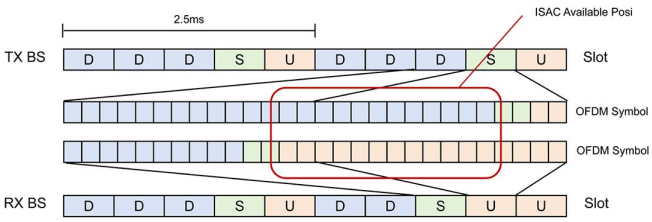


FIG. 15. Radio frame structure design for cooperative sensing system, where D is the downlink slot, S is the special slot and U is the uplink slot.

sensing, the FPGA performs signal preprocessing for the sampled data. L1, representing the physical layer, utilizes sensing-integrated processing units to estimate range, velocity, and angle of the target. The resulting measurements are then transmitted to L2 via a fronthaul application programming interface (FAPI). L2, representing the higher layers, is responsible for calculating the target's location. The SF component handles multinode signal fusion, integrating data from multiple nodes to obtain more accurate sensing results. In terms of communication, the FPGA preprocesses the communication information and then transmits it to L1–L2 for decoding.

Frame Structure and Signal Design

The radio frame structure follows the principles outlined in the 5G standard, ensuring compatibility with existing networks. The transmission and reception windows are configured to align with the cooperative sensing requirements, allowing sufficient time for echo signal acquisition. The prototype supports dynamic allocation of sensing resources within the frame structure, which facilitates adaptive sensing operations. The time-division duplex (TDD) frame structure is shown in Fig. 15, where the transmitting and receiving nodes use different configurations.

Regarding the reference signal design, the prototype adopts optimized signal patterns and base sequences, including the Gold sequence and Zadoff-Chu (ZC) sequence, to achieve high-resolution sensing while maintaining compatibility with communication signals. In communication-centric ISAC, the prototype reuses 5G reference signals to maximize spectral efficiency and minimize network modifications.

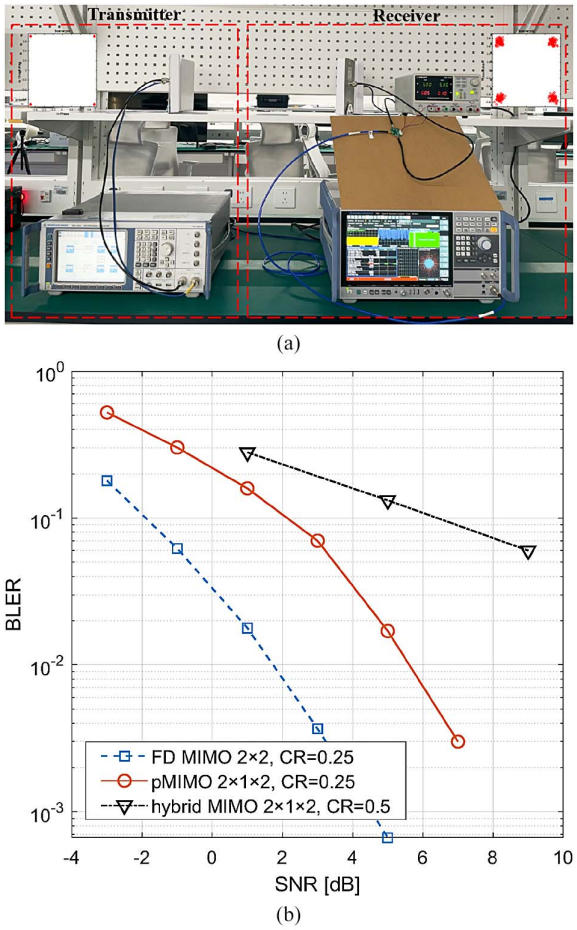


FIG. 16. Pseudo MIMO prototype: (a) structure of the pseudo MIMO prototype and the transmitted and received constellation symbols; and (b) BLER performance of the pseudo MIMO system.

Performance Evaluation

The prototype is planned to be deployed in real-world environments to assess sensing performance under practical conditions. Key performance indicators will include detection probability, and localization accuracy. The evaluation aims to investigate how cooperative sensing improves target detection capabilities compared to single-node sensing, particularly in complex propagation environments.

These experimental validations will provide critical insights for future 6G cooperative sensing system design, highlighting the advantages of leveraging existing 5G infrastructure for ISAC applications.

Prototype of Pseudo MIMO

A basic prototype experiment confirms the effectiveness of the pseudo MIMO transmission scheme, as shown in Fig. 16(a). In this setup, the transmitter uses two antenna elements connected to a Rhode & Schwarz (R&S) signal generator equipped with two RF chains. The receiver consists of two antenna elements connected to an RF device with two input ports; this device reconfigures the analog beamforming patterns. The single output of the device is subsequently fed into an R&S FSW signal and spectrum analyzer, which operates with a single RF chain.

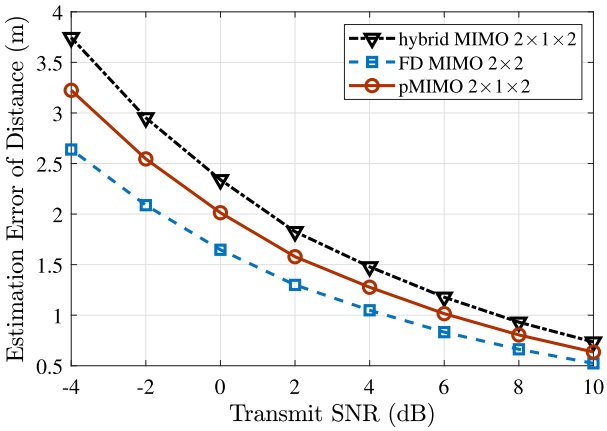


FIG. 17. Estimation error of the range versus the transmit SNR.

The signal generator processes two independently precoded QPSK-modulated baseband streams and upconverts them to the 2.4 GHz band. At the receiver, two distinct analog beamforming patterns switch within the original OFDM sampling period. The RF device combines the received signals, which are then processed by the spectrum analyzer. Experimental results demonstrate that the pseudo MIMO prototype successfully decodes two independent data streams using only one receiving RF chain. Fig. 16(b) presents a comparison of the block error rate (BLER) performance among the pseudo MIMO, hybrid MIMO, and FD MIMO systems. The comparison includes a $2 \times 1 \times 2$ hybrid MIMO configuration that transmits a single data stream and a 2×2 FD-MIMO system that transmits two parallel data streams. Notably, the channel coding rate (CR) for the hybrid MIMO system was doubled to keep the overall code rate consistent. As shown in Fig. 16(b) the pseudo MIMO system achieves a lower BLER than the hybrid MIMO system; however, its BLER remains higher than that of the FD MIMO system due to the use of two fixed, non-optimized analog beamforming patterns.

Furthermore, compared with hybrid MIMO, it can be clearly observed that the pseudo MIMO system increases diversity from one to two as per the slopes of the BLER curves. Fig. 17 presents the estimation error of the range versus the transmit SNR. The estimation error of the range is the product of the speed of light and the estimation error of the time delay (26), which is computed by using the received SNR of the prototype. It can be observed that applying the pseudo MIMO receiver with two fixed analog beamforming patterns can still enhance the sensing accuracy. For example, when the transmit SNR is 4 db, the estimation error of the range is 1.27 and 1.48 m for pseudo MIMO and hybrid MIMO, respectively. Future work will focus on developing a more advanced prototype with dynamically optimized receiving beamforming. In addition, we plan to conduct comprehensive field tests, including the integration of sensing functionality.

CONCLUSION

In this article, we study the cooperative ISAC system for the future 5G-A and 6G network. Starting from the channel model, key technologies including synchronization, beam management and advanced AI-based sensing algorithms are provided. Then the network perspectives of cooperative ISAC system are considered. First, TD and CD methods for interference mitigation are investigated for both UAV and car sensing in 2.6 and 26 GHz bands, respectively. It is found that except for the UAV scenario in the 26 GHz band, CD method outperforms TD method in all cases. Then the static node placement and moving node selection are designed to achieve the optimal sensing signal power and minimal sensing CRB respectively. Moreover, we explore the cooperative ISAC system with novel MIMO technologies including pseudo MIMO and compact MIMO. The key advantage of novel MIMO is that they can improve EE and resulting SNR in sensing, thus enhancing the sensing performance.

Prototypes of cooperative ISAC systems at both Sub 7 GHz and millimeter wave bands are designed for verification. Experimental results show that the system can achieve 0.2 m range accuracy within 500 m range with only 3.5% of the communication capacity decrease.

Overall effectiveness of promising cooperative ISAC systems has been explored and demonstrated in this article. For future work, experimental verifications of cooperative ISAC with multiple sensing nodes, which requires more stringent synchronization as well as advanced data fusion approaches, will be pursued. Large number of sensing targets in the real network requiring more efficient algorithms will be considered. Further refinement of the interference management and node placement as well as selection strategies will be needed. Additional work to verify the performance gain and to assess implementation complexity of novel MIMO technologies in cooperative ISAC system will also be conducted.

REFERENCES

[1] Z. Zhang et al., "6G wireless networks: Vision, requirements, architecture, and key technologies," *IEEE Veh. Technol. Mag.*, vol. 14, no. 3, pp. 28–41, Sep. 2019.

[2] W. Jiang, B. Han, M. A. Habibi, and H. D. Schotten, "The road towards 6G: A comprehensive survey," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 334–366, 2021.

[3] F. Liu et al., "Integrated sensing and communications: Towards dual-functional wireless networks for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, Jun. 2022.

[4] H. Hua, J. Xu, and T. X. Han, "Optimal transmit beamforming for integrated sensing and communication," *IEEE Trans. Veh. Technol.*, vol. 72, no. 8, pp. 10588–10603, Aug. 2023.

[5] X. Wang, Z. Fei, J. A. Zhang, and J. Huang, "Sensing-assisted secure uplink communications with full-duplex BS," *IEEE Commun. Lett.*, vol. 26, no. 2, pp. 249–253, Feb. 2022.

[6] H. Luo, F. Gao, F. Liu, and S. Jin, "6D radar sensing and tracking in monostatic integrated sensing and communications system," 2023, *arXiv:2312.16441*.

[7] S. Li and R. D. Murch, "An investigation into baseband techniques for single-channel full-duplex wireless communication systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 9, pp. 4794–4806, Sep. 2014.

[8] D. Wen and G. Yu, "Time-division cellular networks with full-duplex BSs," *IEEE Commun. Lett.*, vol. 20, no. 2, pp. 392–395, Feb. 2016.

[9] L. Pucci, E. Matricardi, E. Paolini, W. Xu, and A. Giorgetti, "Performance analysis of a bistatic joint sensing and communication system," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Seoul, South Korea, 2022, pp. 73–78.

[10] Z. Han et al., "Multistatic integrated sensing and communication system in cellular networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2023, pp. 123–128.

[11] G. Liu et al., "Cooperative sensing for 6G mobile cellular networks: Feasibility, performance, and field trial," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 10, pp. 2863–2876, Oct. 2024.

[12] L. Xie, S. Song, Y. C. Eldar, and K. B. Letaief, "Collaborative sensing in perceptive mobile networks: Opportunities and challenges," *IEEE Wireless Commun.*, vol. 30, no. 1, pp. 16–23, Feb. 2023.

[13] X. Wang et al., "Cooperative integrated sensing and communication in 6G: From operators perspective," *IEEE Wireless Commun.*, vol. 32, no. 1, pp. 52–59, Feb. 2025.

[14] H. Li, L. Han, R. Duan, and G. M. Garner, "Analysis of the synchronization requirements of 5G and corresponding solutions," *IEEE Commun. Standards Mag.*, vol. 1, no. 1, pp. 52–58, Mar. 2017.

[15] J. Li, G. Zhou, T. Gong, and N. Liu, "A framework for mutual information-based MIMO integrated sensing and communication beamforming design," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 8352–8366, Jun. 2024.

[16] R. Li, Z. Xiao, and Y. Zeng, "Beamforming towards seamless sensing coverage for cellular integrated sensing and communication," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, 2022, pp. 492–497.

[17] X. Wang et al., "Multistatic integrated sensing and communication system based on macro-micro cooperation," *Sensors*, vol. 24, no. 8, 2024.

[18] Y. Niu, Z. Wei, L. Wang, H. Wu, and Z. Feng, "Interference management for integrated sensing and communication systems: A survey," *IEEE Internet Things J.*, vol. 12, no. 7, pp. 8110–8134, Apr. 2025.

[19] S. Wang et al., "Pseudo MIMO (pMIMO): An energy and spectral efficient MIMO-OFDM system," *IEEE Commun. Mag.*, early access, 2025.

[20] Z. Han, Y. Zhang, J. Jin, Q. Wang, and G. Liu, "A novel energy efficient single-RF MIMO antenna combining load modulated array and ESPAR," *IEEE Open J. Antennas Propag.*, early access, 2024.

[21] S. Han, I. C.-L, Z. Xu, and C. Rowell, "Large-scale antenna systems with hybrid analog and digital beamforming for millimeter wave 5G," *IEEE Commun. Mag.*, vol. 53, no. 1, pp. 186–194, Jan. 2015.

[22] Xiaomi, "Summary #3 on ISAC channel modelling," 3rd Generation Partnership Project, Proposal R1-2401496, Feb. 2024. [Online]. Available: https://www.3gpp.org//ftp/TSG_RAN/WG1_RL1/TSGR1_116//docs/

[23] "Study on enhanced LTE support for aerial vehicles," 3rd Generation Partnership Project, Tech. Rep. TR 36.777, Dec. 2017.

[24] "Study on evaluation methodology of new vehicle-to-everything (V2X) use cases for LTE and NR," 3rd Generation Partnership Project, Tech. Rep. TR 37885, Jun. 2019.

[25] "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project, Tech. Rep. TR 38901, Mar. 2024.

[26] G. Liu et al., "Feasibility study of cooperative sensing: Radar cross section, synchronization, cooperative cluster, performance and prototype," *Sci. China Inf. Sci.*, vol. 68, 2025, Art. no. 150302, doi: 10.1007/s11432-024-4377-0.

[27] B. Sundararaman, U. Buy, and A. D. Kshemkalyani, "Clock synchronization for wireless sensor networks: A survey," *Ad Hoc Netw.*, vol. 3, no. 3, pp. 281–323, 2005.

[28] S. B. Chandra Gutha, R. Vinuesa, and H. Azizpour, "Inverse problems with diffusion models: A map estimation perspective," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2025, pp. 4153–4162.

[29] B. Zhu, J. Y. Liu, S. F. Cauley, B. R. Rosen, and M. S. Rosen, "Image reconstruction by domain-transform manifold learning," *Nature*, vol. 555, no. 7697, pp. 487–492, 2018.

[30] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[31] H. K. Aggarwal, M. P. Mani, and M. Jacob, "MoDL: Model-based deep learning architecture for inverse problems," *IEEE Trans. Med. Imag.*, vol. 38, no. 2, pp. 394–405, Feb. 2019.

[32] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.

[33] W. Herzberg, D. B. Rowe, A. Hauptmann, and S. J. Hamilton, "Graph convolutional networks for model-based learning in nonlinear inverse problems," *IEEE Trans. Comput. Imag.*, vol. 7, pp. 1341–1353, 2021.

[34] S. Arridge, P. Maass, O. Aktem, and C.-B. Schanlieb, "Solving inverse problems using data-driven models," *Acta Numerica*, vol. 28, pp. 1–174, 2019, doi: 10.1017/S0962492919000059.
[35] Y. Chen, J. Jin, J. Liu, and Z. Zhao, "Physics-informed neural networks for electromagnetic inverse scattering," *IEEE Trans. Antennas Propag.*, vol. 70, no. 5, pp. 3578–3592, May 2022.
[36] Z. Li, J. Wang, Z. Lin, and L. Zhang, "Transformer for inverse problems in imaging: A review," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, 2024.
[37] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 6840–6851.
[38] H. Chung, Y. Dar, J. Tamir, M. Elad, G. Wetzstein, and H. Seung, "Come-closer-diffuse-faster: Accelerating conditional diffusion models for inverse problems using shrinkage fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 20982–20992.
[39] J. Wang, H. Vardhan, Q. Zhang, and C. Hegde, "Diffusion models for physics-based inverse problems," 2023, *arXiv:2301.10060*.
[40] Z. Han et al., "Microsite placement analysis in 6G multistatic collaborative sensing systems," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2024, pp. 1–6.
[41] G. Liu, L. Han, R. Xi, J. Dong, J. Jin, and Q. Wang, "Sensing accuracy gain, unified performance analysis and optimization in 6G cooperative ISAC systems," *Digit. Commun. Netw.*, 2025, doi: 10.1016/j.dcan.2025.04.007.
[42] T. Ohira and K. Gyoda, "Electronically steerable passive array radiator antennas for low-cost analog adaptive beamforming," in *Proc. IEEE Int. Conf. Phased Array Syst. Technol.*, Piscataway, NJ, USA: IEEE Press, 2000, pp. 101–104.
[43] Z. Han, S. Shen, Y. Zhang, S. Tang, C.-Y. Chiu, and R. Murch, "Single-RF MIMO-OFDM using ESPAR," *IEEE Trans. Veh. Technol.*, vol. 72, no. 5, pp. 6080–6089, May 2023.